OS Orange Problem
(Building PES-VCS)

Himansh Ahuja – PES2UG24CS923

Screenshots:-

PHASE 1:

1A)

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ make test_objects
./test_objects
make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

1B)

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

PHASE 2:

2A)

```
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

2B)

```
6a1af8e11ec4f476
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
```

PHASE 3:

3A)

```
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects
  untracked:  tree.c
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  index.h
  untracked:  pes_debug
  untracked:  README.md
  untracked:  pes.h
  untracked:  pes.c
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  pes_d
  untracked:  commit.h
  untracked:  Makefile
  untracked:  commit.c
  untracked:  tree.h

himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

3B)

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776751915 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776751915 6 file2.txt
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

PHASE 4:

4A)

```
Committed: 02a3deaa806f... Add farewell
commit 02a3deaa806f6aaf3a8cc33a474ea4c494c5f30168774c4a74d093412f219cbd
Author: Himansh Ahuja <PES2UG24CS923>
Date:   1776753574

    Add farewelldd farewelQ

commit d35bc06a03b111dab99cc4d7e47d979f34b530e66e6901b9d9889c56747ab47e
Author: Himansh Ahuja <PES2UG24CS923>
Date:   1776753574

    Add world

commit dff9fa1dea1063d512c6194da352c54115e5b42c2f5f784748c32d5ab8fa3596
Author: Himansh Ahuja <PES2UG24CS923>
Date:   1776753574

    Initial commitver auth!

commit 59c5ba885057930a82ff48b396302ef50535ef58278b98b2f02533b6d2fdcae4
Author: Himansh Ahuja <PES2UG24CS923>
Date:   1776753558

    Initial commitQ

himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

4B)

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/02/a3deaa806f6aaf3a8cc33a474ea4c494c5f30168774c4a74d093412f219cbd
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/59/c5ba885057930a82ff48b396302ef50535ef58278b98b2f02533b6d2fdcae4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d3/5bc06a03b111dab99cc4d7e47d979f34b530e66e6901b9d9889c56747ab47e
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/df/f9fa1dea1063d512c6194da352c54115e5b42c2f5f784748c32d5ab8fa3596
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4C)

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ cat .pes/refs/heads/main
02a3deaa806f6aaf3a8cc33a474ea4c494c5f30168774c4a74d093412f219cbd
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$
```

FINAL:

```
himansh@LAPTOP-90HG81SJ:~/PES2UG24CS923-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
DEBUG: commit_create start
DEBUG: calling tree_from_index
DEBUG: tree_from_index ok
DEBUG: no parent (first commit)
DEBUG: serialize ok
DEBUG: object_write ok
[main c50a2db] Initial commit
Committed: c50a2dbcbf9e... Initial commit

Log after first commit:
commit c50a2dbcbf9ebd336dc3c3226e07ce090fe0c7303506633d0131a21c7c2add2d
Author: Himansh Ahuja <PES2UG24CS923>
Date:   1776753792

    Initial commitQ


--- Second Commit ---
DEBUG: commit_create start
DEBUG: calling tree_from_index
DEBUG: tree_from_index ok
DEBUG: has parent
DEBUG: serialize ok
DEBUG: object_write ok
[main c126d92] Update file.txt
Committed: c126d925ff9d... Update file.txt

--- Third Commit ---
DEBUG: commit_create start
DEBUG: calling tree_from_index
DEBUG: tree_from_index ok
DEBUG: has parent
DEBUG: serialize ok
DEBUG: object_write ok
[main 3327516] Add farewell
Committed: 33275164645c... Add farewell
```